

JUCHENG SHEN

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EDUCATION

Rice University

Expected May 2027

B.A. in Computer Science, Mathematics, and Mathematical Economic Analysis, Major GPA – 3.8/4.0

- Relevant Coursework: Reinforcement Learning, Computer Vision, Honors Linear Algebra, Honors ODE, Real Analysis, Differential Geometry, Advanced Algorithms and Data Structures, Concurrent Programming

RESEARCH EXPERIENCE

California Institute of Technology

May 2026 – Present

Research Intern · Advisor: Prof. Anima Anandkumar

- Investigate how the spectral structure of continuous language representations affects learning efficiency and generalization in diffusion language models.
- Lead the project, develop spectral regularization and attenuation methods, and train and evaluate models on language modeling, length generalization, translation, and summarization.

Rice University

Jan 2026 – Sep 2026

Research Intern · Advisor: Prof. Anastasios Kyrillidis

- Studied how asymmetric input injection enables a shared-weight recurrent Transformer to develop distinct computational roles for reasoning.
- Proposed *Asymmetric Input Recurrence (AIR)* and designed ablation and representation analyses, showing that a clear state-identity signal can induce functional specialization and improve performance on selected abstract reasoning tasks.

Princeton University

Nov 2025 – Feb 2026

Research Intern · Advisor: Prof. Zhuang Liu

- Developed *i1*, a fully open text-to-image model trained from scratch, with systematic exploration of data, architecture, and training choices.
- Built the dataset selection and filtering pipeline and ran 300+ ablation experiments to evaluate modeling and training decisions.

UT Austin & Intel Labs

May 2025 – Sep 2025

Research Intern · Advisors: Prof. Atlas Wang & Dr. Souvik Kundu

- Developed training-free adaptive decoding methods that improve diffusion language model throughput while maintaining generation quality.
- Proposed *OSDT* and led the development of *CadLLM*, designing adaptive decoding strategies and conducting experiments that demonstrated throughput gains of 24–50% and up to $2.3\times$, respectively.

Rice University

Apr 2025 – Aug 2025

Research Intern · Advisor: Prof. Yuke Wang

- Developed *SuperGen*, a system for generating 2–4K videos through low-resolution generation, upscaling, and tiled refinement without fitting the full-resolution video on a single GPU.
- Implemented *SuperGen* on two baselines, conducted end-to-end evaluations and ablation studies, and contributed figures and manuscript text.

PUBLICATIONS

[1] One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer

Jucheng Shen*, Barbara Su*, Anastasios Kyrillidis

NeurIPS 2026

[2] Improving the Throughput of Diffusion-based Large Language Models via a Training-Free Confidence-Aware Calibration

Jucheng Shen, Gaurav Sarkar, Yeonju Ro, Sharath Nittur Sridhar, Zhangyang Wang, Aditya Akella, Souvik Kundu

Findings of ACL 2026

[3] Beyond Static Cutoffs: One-Shot Dynamic Thresholding for Diffusion Language Models

Jucheng Shen, Yeonju Ro

NeurIPS 2025 Efficient Reasoning Workshop

[4] i1: A Simple and Fully Open Recipe for Strong Text-to-Image Models

Boya Zeng, Tianze Luo, Shu Pu, **Jucheng Shen**, Taiming Lu, Gabriel Sarch, Zhuang Liu
NeurIPS 2026 BeNTTo Workshop

PREPRINTS

[5] SuperGen: An Efficient Ultra-High-Resolution Video Generation System with Sketching and Tiling

Fanjiang Ye, Zepeng Zhao, Yi Mu, **Jucheng Shen**, Renjie Li, Kaijian Wang, Saurabh Agarwal, Myungjin Lee, Triston Cao, Aditya Akella, Arvind Krishnamurthy, T. S. Eugene Ng, Zhengzhong Tu, Yuke Wang
arXiv 2025

TEACHING EXPERIENCE

Rice University

Jan 2026 – Present

Teaching Assistant · COMP 282: Computational Optimization of AI

ACADEMIC SERVICE

Reviewer

- NeurIPS 2025 Efficient Reasoning Workshop
- COLM 2026 Efficient Reasoning Workshop

LEADERSHIP EXPERIENCE

HackRice

Apr 2025 – Oct 2025

Board Member (Tracks and Workshops)

- Co-designed and shipped the official starter code used by 500+ hackers, and designed competition tracks and judging criteria for HackRice 15.
- Hosted 2 technical workshops for 100+ attendees; ran Q&A and live coding demos.

Rice Chinese Theater Club

Nov 2023 – Nov 2024

President

- Led a team of 30+ members in organizing semester productions, coordinating venues, licensing, sets and props, publicity, and budgeting.
- Performed in three different plays with a combined audience of over 1,000 across all performances, and represented the club at quarterly meetings of the North American College Chinese Theater Alliance.

HONORS & AWARDS

Frank Liu Jr. Prize for Creative Innovations in Music, Fashion, and the Arts in Napier Rice Launch Challenge 2023.